Size selection (Purification) (1X ratio)

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v1.2, 2026-10-21; 1X bead ratio, 20 µL elution; hand-maintained since 2026-10-21

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Abstract

Size selection (Purification) with 1X ratio to keep more DNA than original version

Procedure

PROCEDURE

1. Calculate the amount of magnetic beads you need and **DISTRIBUTE in a new 1.5 mL eppendorf tube (**Mix the bottle **VIGOROUSLY** to make sure it's homogeneous**).**

For example, for 5 samples distribute 5 × 25 µL = **125 µL** of beads into a new 1.5 mL tube (add about 10% extra for pipetting loss), then dispense 25 µL into each sample.

Sample	Magnetic beads need
1	**25 µl**
2	**50 µl**
3	**75 µl**
4	**100 µl**
5	**125 µl**
6	**150 µl**
7	**175 µl**
8	**200 µl**

Note The ratio is **25 µl of sample** use **25 µl of magnetic beads** to clean up (1 X to keep moderate to long DNA fragments).

Note **Magnetic bead**: BeaverBeads™ DNA Select Isolation

2. In each **25 µL** sample, add **25 µL** magnetic beads in.

Note **Pipetting thoroughly every time** before adding to make sure all magnetic beads mixed well.

3. Mix sample and beads by **gently flicking** the tube for **5 min**.

Note Be gentle, mix too aggressive will make DNA fragmented

4. Transfer the tube to the 8-strip magnetic rack.

After all the magnetic beads are arrested, remove the supernatant.

5. Add **200 µL** 80% ethanol, flip the magnetic rack around to clean the pellet.

Wait until all the magnetic beads are arrested, remove the supernatant.

Note **FRESHLY** make 80% ethanol

6. Repeat the **step 5**.

7. Flash spin the tube, and put on the magnetic rack.

Remove **ANY trace of liquid residuals** using 10 µl pipette.

Note Caution: DO NOT let the beads crack

8. Add **20 µL** elution buffer (17.5 µL of the eluate goes into the EvoPrep end-prep).

Mix gently by flicking the tube and make sure all magnetic beads are dissolved in buffer, then flash spin down the sample.

9. Incubate the tube in PCR thermal cycler using "**37_incubation**" program for **10 min**.

10. Transfer the tube to the magnetic rack.

After all the magnetic beads are arrested, collect the **20 µL** supernatant into a 200 µL PCR tube.

11. The clean-up sample is now ready for 2′ PCR.